Problem sheet 3

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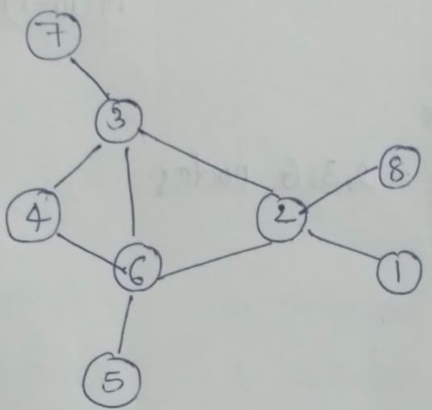
Section-A

Roll no-25

(1)



a) Geodesic distance

b) eccentricity

c) average shortest path

d) radius

e) diameter

f) clustering co-efficient

g) degree centrality

h) Eccentricity centrality

i) Closeness centrality

j) betweness centrality

a)

$$\begin{array}{c|cccccccc|c}
 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & \\
\hline
1 & 0 & 1 & 2 & 3 & 3 & 2 & 3 & 2 & =16 \\
2 & 1 & 0 & 1 & 2 & 2 & 1 & 2 & 1 & 9 \\
3 & 2 & 1 & 0 & 1 & 2 & 1 & 1 & 2 & 7 \\
4 & 3 & 2 & 1 & 0 & 2 & 1 & 2 & 3 & 8 \\
5 & 3 & 2 & 2 & 2 & 0 & 1 & 3 & 3 & 7 \\
6 & 2 & 1 & 1 & 1 & 1 & 0 & 2 & 2 & 4 \\
7 & 3 & 2 & 1 & 2 & 3 & 2 & 0 & 3 & 3 \\
8 & 2 & 1 & 2 & 3 & 3 & 2 & 3 & 0 &
\end{array}$$

b) eccentricity = $\max_j (d(v_i, v_j))$

$e(1) = 3$

$e(2) = 2$

$e(3) = 2$

$e(4) = 3$

$e(5) = 3$

$e(6) = 2$

$e(7) = 3$

$e(8) = 3$

c) avg shortest path $= \dfrac{\sum_{i}^{n} \sum_{j>i}^{n} d(v_i, v_j)}{\binom{n}{2}}$

$= \dfrac{54}{28} = 1.92$

d) radius = 2 (min {e(vi)})

f) clustering coefficient (graph) = 0.208

e) diameter = 3 = max {e(vi)}

$= \frac{2 m_i}{n_i(n_i - 1)}$

g)

1	1
2	4
3	4
4	2
5	1
6	4
7	1
8	1

highest Degree centrality = 2, 3, 6 nodes

h)

1	0.33
2	0.5
3	0.5
4	0.33
5	0.33
6	0.5
7	0.33
8	0.33

Eccentric centrality = 2, 3, 6 nodes

i) $C(v) = \frac{1}{\sum d(u,v)}$

1	0.077
2	0.125
3	0.125
4	0.088
5	0.077
6	0.125
7	0.077
8	0.077

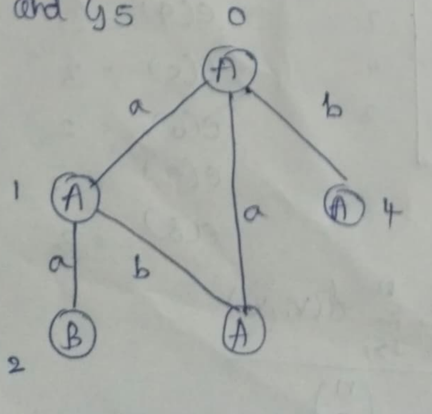
closeness centrality = 2, 3, 6 nodes

(j) Betweeness centrality for 1 ⇒ 0, 2 ⇒ 13.5, 3 ⇒ 8.5, 4 ⇒ 0, 5 ⇒ 0, 6 ⇒ 9.5, 7 ⇒ 0, 8 ⇒ 0

Highest betweeness node = 2

2) G2 and G5

3)



(0, 1, A, a, A)

(1, 2, A, a, B)

(1, 3, A, b, A)

(0, 3, A, a, A)

(0, 4, A, b, A)

DFS order

1. $A \xrightarrow{a} A$
2. $A \xrightarrow{a} B$
3. Backtrack
4. $A \xrightarrow{b} A$
5. backtrack
6. $A \xrightarrow{b} A$

4 1. opinion
2. sentiment
3. sentiment
4. opinion
5. opinion
6. sentiment
7. sentiment
8. sentiment
9. sentiment
10. opinion
11. opinion
12. sentiment
13. sentiment